



VNL uses a cascading star network architecture to enable rural wireless telephony and Internet connectivity (Source: VNL)

operate in the low-Earth orbit. Though there has been no official confirmation, it is rumoured that Apple will be a potential investor-partner in this venture.

Elon Musk's company SpaceX is also planning to plant more than 4000 Internet-providing satellites around Earth between 2019 and 2024. After testing a few satellites this year, the company will start manufacturing and launching its satellites into the low-Earth orbit using its own Falcon 9 rockets. The satellites will operate in Ka- and Ku-band frequencies. Placing the satellites at low altitudes of 1110km and 1325km in the low-Earth orbit will help the company save cost and time.

Some experts fear that such a large number of satellites would increase the amount of space junk and make it difficult for companies hoping to place satellites in that orbit in the future. However, the company claims that this will make it easier for everybody to get Internet access, and also the network will be able to adapt to needs, allocating resources to specific areas during busy times and avoiding interference with other systems.

OneWeb hopes to differentiate itself with the affordability and speed at which it can manufacture satellites. The company claims to be building the world's fastest satellite production facility, which will mass produce satellites just like medical or avionics equipment are manufactured today. Like SpaceX, OneWeb is also targeting the low-Earth orbit.

Last year, the Federal Communications Commission (FCC) approved OneWeb's proposal to place its non-geostationary satellites at an altitude of about 1200km. Due to this, signals coming from the satellite will have significantly less latency of around 30ms, resulting in much faster communications. Compare this with the 35000+ km altitude for satellites used by traditional Internet service providers (ISPs), causing a latency of 600ms or greater. Starting this year, OneWeb plans to start launching over 700 satellites in association with Airbus. The constellation of satellites will logically interlock to cover the entire planet.

On the Earth, there will be low-cost user terminals that connect to the satellites, and provide long-term evolution (LTE), 3G and Wi-Fi connectivity to surrounding areas. OneWeb hopes to tie-up with existing ISPs and mobile operators to extend their networks into remote rural areas. According to the company, its small cell terminals and core network will be fully 3GPP compatible, so partner operators will be able to use its infrastructure with their current customers, devices and billing systems.

In a news report, FCC Chairman Ajit Pai commented in favour of such satellite constellations: "We hope to approve many more constellations because we know that the more the companies compete, the more consumers win. Additionally, the commission has an ongoing rulemaking proceeding, proposing to update the

current NGSO Fixed Satellite Service rules to better accommodate this next generation of systems."

Delivered through balloons

Google is attempting to provide Internet connectivity to remote areas using high-flying balloons. The idea behind Project Loon, which is part of Google's Alphabet X research labs, is to suspend a set of huge balloons in the stratosphere at an altitude of around 20km. These balloons will contain the infrastructure needed to provide connectivity up to about 80km by using LTE wireless communications. Loon will share cellular spectrum with telecommunications companies. When users in remote areas access the Internet using their cell phones or other LTE-enabled devices, balloons will relay wireless traffic from these devices back to the global Internet using high-speed links. As of last year, the prototypes provided 15Mbps Internet access.

Google is still working to perfect the way balloons navigate. The stratosphere is made of layers, each having a different wind speed and direction. By moving the balloons up and down to different layers, Google team can make these move to the required region. By coordinating positions of different balloons, they can make balloons work together to cover a region.

According to a recent press report, Project Loon now uses machine learning to predict weather systems and gain greater control over the movement of balloons. It is now possible to focus on a specific region for a long period of time. So far the tests are promising, but the company has still not spoken about a commercial rollout.

In the meanwhile, the Andhra Pradesh government has realised that if not the balloons, the underlying technology can be put to good use right away. The technology being used by Loon is known as free space optical communications (FSOC), wherein beams of light are used to deliver high-speed, high-capacity connectivity over long distances.

AP State FiberNet, a telecom provider in Andhra Pradesh has now entered into an agreement with Alpha-